

USENIX Security '26 Cycle 1

Accepted Papers

USENIX Security '26 has two submission deadlines. Prepublication versions of the accepted papers from the first submission deadline are available below.

VIEW MODE: [CONDENSED](#) | [STANDARD](#) | [EXPANDED](#)

[Hide details](#) ▾

[Bond: Constraint-Directed Fuzzing for Automated Validation of Taint Analysis Results in Linux-based IoT Firmware](#)

Jiaqian Peng, *Institute of Information Engineering, Chinese Academy of Sciences; School of Cyber Security, University of Chinese Academy of Sciences*; Puzhuo Liu, *Ant Group; Tsinghua University*; Kai Cheng, *Institute of Information Engineering, Chinese Academy of Sciences*; Zhaoteng Yan, *School of Cyber Security, University of Chinese Academy of Sciences*; Jie Liu, *Institute of Information Engineering, Chinese Academy of Sciences*; Chengnian Sun, *University of Waterloo*; Hongsong Zhu, *Institute of Information Engineering, Chinese Academy of Sciences; School of Cyber Security, University of Chinese Academy of Sciences*

AVAILABLE MEDIA 

[Show details](#) ►

[Heli: Heavy-Light Private Aggregation](#)

Ryan Lehmkuhl and Henry Corrigan-Gibbs, *MIT*; Emma Dauterman, *Stanford*; David J. Wu, *University of Texas at Austin*

AVAILABLE MEDIA 

[Show details](#) ►

["Your imaging may be stone-cold normal, but if they look sick, they're going to get admitted": An Investigation of Clinicians' Perceptions of Impact & Likelihood of Security Failures](#)

Ronald E. Thompson III and Hamza Khalid, *Tufts University*; Hilary Fisher, *Brigham & Women's Hospital*; Rhea Votipka, *Beth Israel Lahey Health*; Daniel Votipka, *Tufts University*

AVAILABLE MEDIA 

[Show details](#) ►

[Hop: A Modern Transport and Remote Access Protocol](#)

Paul Flammarion, *Stanford University*; George Hosono, *Georgia Institute of Technology*; Wilson Nguyen, Laura Bauman, Daniel Rebelsky, and Gerry Wan, *Stanford University*; David Adrian,

Independent; Zakir Durumeric, Stanford University

AVAILABLE MEDIA 

[Show details](#) ▶

Interpolation-Based Optimization for Enforcing ℓ_p -Norm Metric Differential Privacy in Continuous and Fine-Grained Domains

Chenxi Qiu, *University of North Texas*

AVAILABLE MEDIA 

[Show details](#) ▶

Membership Inference Attacks on Tokenizers of Large Language Models

Meng Tong, *University of Science and Technology of China*; Yuntao Du, *Purdue University*; Kejiang Chen and Weiming Zhang, *University of Science and Technology of China*; Ninghui Li, *Purdue University*

AVAILABLE MEDIA 

[Show details](#) ▶

JailbreakScope: Interpreting Jailbreak Mechanism through Representation and Circuit Analyses

Zeqing He, Zhibo Wang, Zhixuan Chu, Huiyu Xu, Wenhui Zhang, Qinglong Wang, and Rui Zheng, *Zhejiang University*

AVAILABLE MEDIA 

[Show details](#) ▶

Imitative Membership Inference Attack

Yuntao Du and Yuetian Chen, *Purdue University*; Hanshen Xiao, *Purdue University & NVIDIA Research*; Bruno Ribeiro and Ninghui Li, *Purdue University*

AVAILABLE MEDIA 

[Show details](#) ▶

SMASH: Scalable Maliciously Secure Hybrid Multi-party Computation Framework for Privacy-Preserving Large Language Models

Yunlv Lv and Rui Zhang, *Institute of Information Engineering, Chinese Academy of Sciences; State Key Laboratory of Cyberspace Security Defense; School of Cyber Security, University of Chinese Academy of Sciences*; Zhiyuan Zhang, *Max Planck Institute for Security and Privacy*; Ziyi Wan, *Institute of Information Engineering, Chinese Academy of Sciences; State Key Laboratory of Cyberspace Security Defense; School of Cyber Security, University of Chinese Academy of Sciences*; Lanxue Zhang, *Institute of Information Engineering, Chinese Academy of Sciences; School of Cyber Security, University of Chinese Academy of Sciences*; Minhui Xue, *CSIRO's Data61 and Responsible AI Research (RAIR) Centre, Adelaide University*; Jiangtao Li, *East China Normal University*; Yanan Cao, *Institute of Information Engineering, Chinese Academy of Sciences; School of Cyber Security, University of Chinese Academy of Sciences*

AVAILABLE MEDIA 

[Show details](#) ▶

XGuardian: Towards Generalized, Explainable and More Effective Server-side Anti-cheat in First-Person Shooter Games

Jiayi Zhang, Chenxin Sun, and Chenxiong Qian, *The University of Hong Kong*

AVAILABLE MEDIA 

[Show details](#) ▶

The Prompt Stealing Fallacy: Rethinking Metrics, Attacks, and Defenses

Zehang Deng, *Swinburne University of Technology and CSIRO's Data61*; Haoyang Li, *The Hong Kong Polytechnic University*; Wanlun Ma, *Swinburne University of Technology*; Ruoxi Sun and Derui Wang, *CSIRO's Data61*; Minhui Xue, *CSIRO's Data61 and Responsible AI Research (RAIR) Centre, Adelaide University*; Haibo Hu, *The Hong Kong Polytechnic University*; Sheng Wen and Yang Xiang, *Swinburne University of Technology*

AVAILABLE MEDIA 

[Show details](#) ▶

When AIOps Become "AI Oops": Subverting LLM-driven IT Operations via Telemetry Manipulation

Dario Pasquini, *RSAC Labs*; Evgenios M. Kornaropoulos and Giuseppe Ateniese, *George Mason University*; Omer Akgul, Athanasios Theocharis, and Petros Efstathopoulos, *RSAC Labs*

AVAILABLE MEDIA 

[Show details](#) ▶

A Large-Scale Study of Personalized Phishing using Large Language Models

Stefan Czybik, *BIFOLD & TU Berlin*; Anne Josiane Kouam, *Inria & TU Berlin*; Peter Heubl and Jan Magnus Nold, *Ruhr-University Bochum*; Konrad Rieck, *BIFOLD & TU Berlin*

AVAILABLE MEDIA 

[Show details](#) ▶

Secure Protocol Composition under Dynamic Corruption: Scaling Up Symbolic Analysis for Real-World Security Properties

Cas Cremers, Erik Pallas, and Aleksi Peltonen, *CISPA Helmholtz Center for Information Security*

AVAILABLE MEDIA 

[Show details](#) ▶

Opossum Attack: Application Layer Desynchronization using Opportunistic TLS

Robert Merget, *Technology Innovation Institute*; Nurullah Erinola and Marcel Maehren, *Ruhr University Bochum*; Lukas Knittel, *Ruhr University Bochum*; Sven Hebrok, *Paderborn University*; Marcus Brinkmann, *Ruhr University Bochum*; Juraj Somorovsky, *Paderborn University*; Jörg Schwenk, *Ruhr University Bochum*

AVAILABLE MEDIA 

[Show details](#) ▶

Breaking Widely Deployed Perceptual Hash Functions: Black-Box Collisions in Apple NeuralHash and Microsoft PhotoDNA

Diane Leblanc-Albarel and Bart Preneel, *KU Leuven*

AVAILABLE MEDIA 

[Show details](#) ▶

Zero Knowledge (About) Encryption: A Comparative Security Analysis of Three Cloud-based Password Managers

Matteo Scarlata, *ETH Zurich*; Giovanni Torrisi and Matilda Backendal, *USI, Lugano*; Kenneth G. Paterson, *ETH Zurich*

AVAILABLE MEDIA 

[Show details ▶](#)

A Midsummer Meme's Dream: Investigating Market Manipulations in the Meme Coin Ecosystem

Alberto Maria Mongardini, *Sapienza University of Rome and Technical University of Denmark*; Alessandro Mei, *Sapienza University of Rome*

AVAILABLE MEDIA 

[Show details ▶](#)

Distributed Vector Commitments and Their Applications

Rui Gao and Huaqun Wang, *Nanjing University of Posts and Telecommunications, The State Key Laboratory of Tibetan Intelligence*; Zhiguo Wan, *Hangzhou Normal University*; Yuncong Hu, *Shanghai Jiao Tong University*

AVAILABLE MEDIA 

[Show details ▶](#)

From Mirai to Gorilla: Deep Dive into a Long-Lasting DDoS-for-Hire Botnet

Maarten Weyns, Dario Ferrero, and Stefan Op de Beek, *Delft University of Technology (TU Delft)*; Daniel Wagner, *Max Planck Institute for Informatics / DE-CIX*; Georgios Smaragdakis and Harm Griffioen, *Delft University of Technology (TU Delft)*

AVAILABLE MEDIA 

[Show details ▶](#)

LPG: Raise Your Location Privacy Game in Direct-to-Cell LEO Satellite Networks

Quan Shi, *National University of Singapore*; Liying Wang, *Peking University*; Prosanta Gope, *The University of Sheffield*; Qi Liang and Haowen Wang, *Beijing University of Posts and Telecommunications*; Qirui Liu and Chenren Xu, *Peking University*; Shangguang Wang and Qing Li, *Beijing University of Posts and Telecommunications*; Biplab Sikdar, *National University of Singapore*

AVAILABLE MEDIA 

[Show details ▶](#)

PrivacyShield: Relaying BLE Beacons to Counter Unsolicited Tracking

Florian Hofhammer, *EPFL*; Daniele Antonioli, *EURECOM*; Mathias Payer, *EPFL*

AVAILABLE MEDIA 

[Show details ▶](#)

Assessing LLM Response Quality in the Context of Technology-Facilitated Abuse

Vijay Prakash, *New York University*; Majed Almansoori, *University of Wisconsin–Madison*; Donghan Hu, *New York University*; Rahul Chatterjee, *University of Wisconsin–Madison*; Danny Yuxing Huang, *New York University*

AVAILABLE MEDIA 

[Show details ▶](#)

Residual-PAC Privacy: Automatic Privacy Control Beyond the Gaussian Barrier

Tao Zhang and Yevgeniy Vorobeychik, *Washington University in St. Louis*

AVAILABLE MEDIA 

[Show details ▶](#)

HAMLOCK: HARDware-Model LOGically Combined attack

Sanskar Amgain, *University of Tennessee, Knoxville*; Daniel Lobo, Atri Chatterjee, and Swarup Bhunia, *University of Florida*; Fnu Suya, *University of Tennessee, Knoxville*

AVAILABLE MEDIA 

[Show details ▶](#)

Verity: Verifiable Local Differential Privacy

James Bell-Clark, Adrià Gascón, Baiyu Li, and Mariana Raykova, *Google*; Amrita Roy Chowdhury, *University of Michigan*

AVAILABLE MEDIA 

[Show details ▶](#)

Missing, Present and Conflicting: A Large Scale Analysis of IoT Update Information in the EU Market

Swaathi Vetrivel, Michel van Eeten, and Carlos H. Gañán, *Delft University of Technology*

AVAILABLE MEDIA 

[Show details ▶](#)

VSG-Safe: Spotting NSFW Video through Cross-Frame Evidence

Yuyang Zhang, Xudong Jiang, Yuxuan Song, and Yuxiang Sun, *Wuhan University*; Yihao Huang, *National University of Singapore*; Run Wang, Shundi Xiao, and Lina Wang, *Wuhan University*

AVAILABLE MEDIA 

[Show details ▶](#)

Cutting the Gordian Knot: Detecting Malicious PyPI Packages via a Knowledge-Mining Framework

Wenbo Guo, *Nanyang Technological University*; Chengwei Liu, *Nankai University*; Ming Kang, *Sichuan University*; Yiran Zhang and Jiahui Wu, *Nanyang Technological University*; Zhengzi Xu, *Imperial Global Singapore*; Vinay Sachidananda and Yang Liu, *Nanyang Technological University*

AVAILABLE MEDIA 

[Show details ▶](#)

End-to-End Encrypted Collaborative Documents

Christian Knabenhans, *EPFL*; Zayd Maradni, *Max Planck Institute for Software Systems*; Carmela Troncoso, *Max Planck Institute for Security and Privacy & EPFL*

AVAILABLE MEDIA 

[Show details ▶](#)

SophOMR: Improved Oblivious Message Retrieval from SIMD-Aware Homomorphic Compression

Keewoo Lee, *Ethereum Foundation*; Yongdong Yeo, *Seoul National University*

AVAILABLE MEDIA 

[Show details ▶](#)

FABS: Fast Attribute-Based Signatures

Liqun Chen, Long Meng, Yalan Wang, Nada El Kassem, Christopher JP Newton, Yangguang Tian, Jodie Knapp, Constantin Cătălin Drăgan, and Daniel Gardham, *University of Surrey*; Mark Manulis, *Universität der Bundeswehr München*

AVAILABLE MEDIA 

[Show details ▶](#)

Invariant-Guided Logical Testing of Open RAN Controllers

Tianchang Yang, Ali Ranjbar, Gang Tan, and Syed Rafiul Hussain, *The Pennsylvania State University*

AVAILABLE MEDIA 

[Show details ▶](#)

Analyzing the WebRTC Ecosystem and Breaking Authentication in DTLS-SRTP

Martin Bach, *Technology Innovation Institute*; Vukašin Karadžić, *TU Darmstadt*; Lukas Knittel, *Ruhr-University Bochum*; Robert Merget and Jean Paul Degabriele, *Technology Innovation Institute*

AVAILABLE MEDIA 

[Show details ▶](#)

Security and Privacy Analysis of Tile's Location Tracking Protocol

Akshaya Kumar, Anna Raymaker, and Michael A. Specter, *Georgia Institute of Technology*

AVAILABLE MEDIA 

[Show details ▶](#)

VIPER Strike: Defeating Visual Reasoning CAPTCHAs via Structured Vision–Language Inference

Minfeng Qi and Dongyang He, *City University of Macau*; Qin Wang, *CSIRO Data61*; Lefeng Zhang, *City University of Macau*

AVAILABLE MEDIA 

[Show details ▶](#)

Stayin' Alive: How Global Stolen Data Markets Thrive on Telegram

Tina Marjanov, *University of Cambridge*; Taro Tsuchiya, *Carnegie Mellon University*; Konstantinos Ioannidis and Jack Hughes, *University of Cambridge*; Nicolas Christin, *Carnegie Mellon University*; Alice Hutchings, *University of Cambridge*

AVAILABLE MEDIA 

[Show details ▶](#)

Assumption-Free Fuzzy PSI via Predicate Encryption

Erik-Oliver Blass, *Airbus*; Guevara Noubir, *Northeastern University*

AVAILABLE MEDIA 

[Show details ▶](#)

Logos: Robust Sharding Blockchain With Fast Processing and Optimal Cross-Shard Overhead

Yizhong Liu, *Beihang University and Beijing Advanced Innovation Center for Future Blockchain and Privacy Computing*; Boyu Zhao, Yuxuan Hu, Haojun Tan, Feiang Ran, Andi Liu, and Zhuocheng Pan, *Beihang University*; Yuan Lu, *Institute of Software, Chinese Academy of Sciences*; Song Bian, Jianwei Liu, and Zhenyu Guan, *Beihang University*

AVAILABLE MEDIA 

[Show details](#) ▶

SoK: PHILTER: Uncovering Security and Functional Gaps in AI-based Phishing Website Detection Literature via an LLM-based Reasoning Framework

Mahbub Alam, *Texas A&M University*; Muhammad Lutfor Rahman, *California State University San Marcos*; Sonjoy Kumar Paul, Amy W. Hays, Aftab Hussain, Md Imanul Huq, and Nitesh Saxena, *Texas A&M University*

AVAILABLE MEDIA 

[Show details](#) ▶

Streaming Function Secret Sharing and Its Applications

Xiangfu Song, *Nanyang Technological University*; Jianli Bai, *Singapore Management University*; Ye Dong, *National University of Singapore*; Yijian Liu, Yu Zhang, and Xianhui Lu, *Institute of Information Engineering, Chinese Academy of Sciences and School of Cyber Security, University of Chinese Academy of Sciences*; Tianwei Zhang, *Nanyang Technological University*

AVAILABLE MEDIA 

[Show details](#) ▶

Shred-to-Shine Metamorphosis of (Distributed) Polynomial Commitments

Weihan Li, *School of Cyber Science and Technology, Beihang University*; Ant Group; Zongyang Zhang, *School of Cyber Science and Technology, Beihang University*; Sherman S. M. Chow, *The Chinese University of Hong Kong*; Yanpei Guo, *National University of Singapore*; Boyuan Gao, *School of Cyber Science and Technology, Beihang University*; Xuyang Song, *Anoma*; Yi Deng, *School of Cryptology, Xidian University*; Jianwei Liu, *School of Cyber Science and Technology, Beihang University*

AVAILABLE MEDIA 

[Show details](#) ▶

Unbalanced Fuzzy Private Set Intersection for L_∞ Distance: Achieving Sublinear Communication with Large Set Size

Shengzhe Meng and Xiaodong Wang, *Tsinghua University*; Xv Zhou, *Beihang University*; Bei Liang, *Beijing Institute of Mathematical Sciences and Applications*

AVAILABLE MEDIA 

[Show details](#) ▶

mmCipher: Batching Post-Quantum Public Key Encryption Made Bandwidth-Optimal

Hongxiao Wang, *The University of Hong Kong*; Ron Steinfeld, *Monash University*; Markku-Juhani O. Saarinen, *Information Security Laboratory, Tampere University*; Muhammed F. Esgin, *Monash University*; Siu-Ming Yiu, *The University of Hong Kong*

AVAILABLE MEDIA 

[Show details ▶](#)

DNS Cache Poisoning Like it's 2006

Omer Ben-Simhon and Amit Klein, *Hebrew University of Jerusalem*

AVAILABLE MEDIA 

[Show details ▶](#)

UncoreBleed: AEX-Free, High-Resolution, and Low-Noise Side-Channel Attacks on SGX Enclaved Execution

Decheng Chen, *South China University of Technology*; Zhi Zhang, *The University of Western Australia*; Zhenkai Zhang, *Clemson University*; Xin Zhang, *Shandong University*; Yansong Gao, *Southeast University*; Yi Zou, *South China University of Technology*

AVAILABLE MEDIA 

[Show details ▶](#)

M-Step: A Single-Stepping Framework for Side-Channel Analysis on TrustZone-M

Cristiano Rodrigues, *Centro ALGORITMI, Universidade do Minho*; Marton Bognar, *DistriNet, KU Leuven*; Sandro Pinto, *Centro ALGORITMI, Universidade do Minho*; Jo Van Bulck, *DistriNet, KU Leuven*

AVAILABLE MEDIA 

[Show details ▶](#)

DaLens: Charting DNS Self-Amplification Threats at Large

Liwen Xu and Zechao Cai, *ETH Zurich*; Huayi Duan, *HKUST(GZ)*; Adrian Perrig, *ETH Zurich*

AVAILABLE MEDIA 

[Show details ▶](#)

Inconsistent, Incomplete, and Insecure: A Survey of Account Security Interfaces

Arkaprabha Bhattacharya and Alaa Daffalla, *Cornell University*; Kevin Lee, *Independent Researcher*; Rosanna Bellini, *New York University*; Nicola Dell, *Cornell Tech*; Thomas Ristenpart, *University of Toronto & Cornell Tech*

AVAILABLE MEDIA 

[Show details ▶](#)

Hydrangea: Optimistic Two-Round Partial Synchrony with Improved Fault Resilience

Nibesh Shrestha, *Supra Research*; Aniket Kate, *Supra Research / Purdue University*; Kartik Nayak, *Duke University*

AVAILABLE MEDIA 

[Show details ▶](#)

Revealing the Dark Side of Smart Accounts: An Empirical Study of EIP-7702 Incurred Risks in Blockchain Ecosystem

Mingyuan Huang, *Hong Kong University of Science and Technology*; Han Liu, *College of Cryptology and Cyber Science, Nankai University*; Shuo Yang, *Sun Yat-sen University*; Daoyuan Wu, *Lingnan University*; Shuai Wang, *Hong Kong University of Science and Technology*

AVAILABLE MEDIA 

[Show details ▶](#)

DMGuard: Safeguarding Kernels from Physical-Page Use-After-Free Vulnerabilities

Juhee Kim, Jaeyoung Chung, Dae R. Jeong, and Byoungyoung Lee, *Seoul National University*

Show details ▶

Garuda and Pari: Faster and Smaller SNARKs via Equiefficient Polynomial Commitments

Michel Dellepere, *Independent*; Pratyush Mishra and Alireza Shirzad, *University of Pennsylvania*

AVAILABLE MEDIA 

Show details ▶

Transparent Dictionaries from Polynomial Commitments

Hossein Hafezi, *NYU*; Alireza Shirzad, *University of Pennsylvania*; Benedikt Bünz and Joseph Bonneau, *NYU*

AVAILABLE MEDIA 

Show details ▶

Efficient Threshold ML-DSA

Sofía Celi, *Brave Research*; Rafael del Pino and Thomas Espitau, *PQShield*; Guilhem Niot, *PQShield and Univ Rennes, CNRS, IRISA*; Thomas Prest, *PQShield*

AVAILABLE MEDIA 

Show details ▶

Nudge: A Private Recommendations Engine

Alexandra Henzinger, *MIT*; Emma Dauterman, *MIT and Stanford*; Henry Corrigan-Gibbs, *MIT*; Dan Boneh, *Stanford*

AVAILABLE MEDIA 

Show details ▶

On Evaluating the Robustness of Large Vision-Language Models via Untargeted Modality Alignment Breaking Adversarial Attack

Zhichao Li, Hongshan Yang, Zhibo Wang, Huiyu Xu, and Junhong Lai, *Zhejiang University*; Yaopeng Wang, *Southeastern University and Zhejiang University*; Kui Ren and Chun Chen, *Zhejiang University*

AVAILABLE MEDIA 

Show details ▶

From Easy to Hard++: Promoting Differentially Private Image Synthesis Through Spatial-Frequency Curriculum

Chen GONG and Kecen Li, *University of Virginia*; Zinan Lin, *Microsoft Research*; Tianhao Wang, *University of Virginia*

AVAILABLE MEDIA 

Show details ▶

FIRA: Enabling Automatic Forensic Investigation of Unmanned Aerial Vehicles

Yizhi Huang, *Georgia Institute of Technology*; David Oygenblik, *Georgia Tech*; Runze Zhang, Mingxuan Yao, Muhammad Ibrahim, Burak Sahin, Haichuan Xu, Saman Zonouz, and Brendan Saltaformaggio, *Georgia Institute of Technology*

AVAILABLE MEDIA 

[Show details ▶](#)

MASLeak: Investigating and Exposing Intellectual Property Leakage Vulnerabilities in Multi-Agent Systems

Liwen Wang, *The Hong Kong University of Science and Technology*; Wenxuan Wang, *Renmin University of China*; Shuai Wang, Zongjie Li, Zhenlan Ji, and Zongyi LYU, *The Hong Kong University of Science and Technology*; Daoyuan Wu, *Lingnan University*; Shing-Chi Cheung, *The Hong Kong University of Science and Technology*

AVAILABLE MEDIA 

[Show details ▶](#)

Love, Lies, and Language Models: Investigating AI's Role in Romance-Baiting Scams

Gilad Gressel and Rahul Pankajakshan, *Center for Cybersecurity Systems & Networks, Amrita Vishwa Vidyapeetham, Amritapuri*; Shir Rozenfeld, *Ben Gurion University of the Negev*; Ling Li, *Ca' Foscari University of Venice*; Ivan Franceschini, *University of Melbourne*; Krishnashree Achuthan, *Center for Cybersecurity Systems & Networks, Amrita Vishwa Vidyapeetham, Amritapuri*; Yisroel Mirsky, *Ben Gurion University of the Negev*

AVAILABLE MEDIA 

[Show details ▶](#)

United We Defend: Collaborative Membership Inference Defenses in Federated Learning

Li Bai, Junxu Liu, Sen Zhang, Xinwei Zhang, Qingqing Ye, and Haibo Hu, *The Hong Kong Polytechnic University*

AVAILABLE MEDIA 

[Show details ▶](#)

Tracegram: Framing Trace-Level Traffic Analysis with Temporally-Aware Multiple Instance Learning

Jian Qu, Yuchen Zhang, Jialong Zhang, Jianfeng Li, and Xiaobo Ma, *School of Computer Science and Technology, Xi'an Jiaotong University*

AVAILABLE MEDIA 

[Show details ▶](#)

"Abuse Risks are Often Inherent to Product Features": Exploring AI Vendors' Bug Bounty and Responsible Disclosure Policies

Yangheran Piao, Jingjie Li, and Daniel W. Woods, *University of Edinburgh*

AVAILABLE MEDIA 

[Show details ▶](#)

PICS: Private Intersection over Committed (and reusable) Sets

Aarushi Goel, *Rutgers University*; Peihan Miao and Phuoc Van Long Pham, *Brown University*; Satvinder Singh, *Purdue University*

AVAILABLE MEDIA 

[Show details ▶](#)

MULCOTAINT: Towards Efficient Multi-tag Dynamic Taint Analysis via Hardware/Software Co-design

Bing Qi, *University of Chinese Academy of Sciences; Institute of Software, Chinese Academy of Sciences*; Yi Yang and Xiangkun Jia, *University of Chinese Academy of Sciences; Institute of Software, Chinese Academy of Sciences; Key Laboratory of System Software (Chinese Academy of Sciences)*; Zhengpin Qian and Huafeng Huang, *University of Chinese Academy of Sciences; Institute of Software, Chinese Academy of Sciences*; Purui Su, *University of Chinese Academy of Sciences; Institute of Software, Chinese Academy of Sciences; Key Laboratory of System Software (Chinese Academy of Sciences)*

AVAILABLE MEDIA 

[Show details ▶](#)

WAVED: Principled Identification of Off-Path Exploitable Weak Verifications within the TCP/IP Protocol Suite

Yizhou Zhao and Xuewei Feng, *Tsinghua University*; Min Li, *Zhongguancun Laboratory*; Ke Xu, *Tsinghua University*

AVAILABLE MEDIA 

[Show details ▶](#)

CuSafe: Capturing Memory Corruption on NVIDIA GPUs

Hongyi Lu, *Southern University of Science and Technology and Hong Kong University of Science and Technology*; Fengwei Zhang, *Southern University of Science and Technology*; Zhenkai Zhang, *Clemson University*; Shuai Wang, *Hong Kong University of Science and Technology*; Yanan Guo, *University of Rochester*

AVAILABLE MEDIA 

[Show details ▶](#)

Attacks on Approximate Caches in Text-to-Image Diffusion Models

Desen Sun, Shuncheng Jie, and Sihang Liu, *University of Waterloo*

AVAILABLE MEDIA 

[Show details ▶](#)

Side-Channel Attacks on Open vSwitch

Daewoo Kim and Sihang Liu, *University of Waterloo*

AVAILABLE MEDIA 

[Show details ▶](#)

A Distortion-minimization Watermarking Framework for Large Language Models: Larger Capacity, Stronger Robustness and Higher Quality

Liming Zhai, Xuezhou Shang, Liyun Zhang, and Po Hu, *Central China Normal University*

AVAILABLE MEDIA 

[Show details ▶](#)

Behind Bars: A Side-Channel Attack on NVIDIA MIG Cache Partitioning Using Memory Barriers

Cheng Gu, *University of Rochester*; Reese Levine, *UC Santa Cruz*; Zhenkai Zhang, *Clemson University*; Tyler Sorensen, *Microsoft and UC Santa Cruz*; Yanan Guo, *University of Rochester*

AVAILABLE MEDIA 

[Show details ▶](#)

TrojPix: Electromagnetic Covert Channels via Imperceptible Pixel Modulation

Guoming Zhang, *Shandong University and Quan Cheng Laboratory*; Huiting Zhang, Zhenwei Lu, Heqiang Fu, Xin Gao, Riccardo Spolaor, and Yetong Cao, *Shandong University*; Yanni Yang and Pengfei Hu, *Shandong University and Quan Cheng Laboratory*

AVAILABLE MEDIA 

[Show details](#) ▶

Anonymous Tokens with Designated-Reader Metadata Bit

Aisha Tu, *Wuhan University*; Meng Jia, *The Hong Kong Polytechnic University*; Kun He, Jing Chen, and Ruiying Du, *Wuhan University*

AVAILABLE MEDIA 

[Show details](#) ▶

Paper Title Under Embargo

Author list under embargo.

[Show details](#) ▶

TIMESLICE-SANDWICH: A GPU Side-Channel Attack Exploiting Time-Sliced Scheduling

Hodong Kim and Gyeongsup Lim, *Korea University*; Seunghee Shin, *State University of New York at Binghamton*; Youngjoo Shin and Junbeom Hur, *Korea University*

AVAILABLE MEDIA 

[Show details](#) ▶

Bridging Usability and Performance: A Tensor Compiler for Autovectorizing Homomorphic Encryption

Edward Chen, Fraser Brown, and Wenting Zheng, *Carnegie Mellon University*

AVAILABLE MEDIA 

[Show details](#) ▶

Shadowfax: Hybrid Security and Deniability for AKEMs

Phillip Gajland, *IBM Research Europe – Zurich*; Vincent Hwang, *MPI-SP, Radboud University*; Jonas Janneck, *Ruhr University Bochum*

AVAILABLE MEDIA 

[Show details](#) ▶

RBOOT: Accelerating Homomorphic Neural Network Inference by Fusing ReLU within Bootstrapping

Zhaomin Yang, Chao Niu, Benqiang Wei, Zhicong Huang, Cheng Hong, and Tao Wei, *Ant Group*

AVAILABLE MEDIA 

[Show details](#) ▶

PANGOLIN: Fuzzing Multilingual IoT Firmware with LLM-Driven Code Analysis

Zhipeng Jia and Xiaokang Yin, *Information Engineering University*; Shuitao Gan, *Laboratory for Advanced Computing and Intelligence Engineering*; Chao Zhang, *Institute for Network Sciences and Cyberspace, Tsinghua University*; JCSS, *Tsinghua University (INSC) - Science City (Guangzhou) Digital Technology Group Co., Ltd.*; Hangtian Liu, *State Key Laboratory of Mathematical Engineering and Advanced Computing*; Jiangnan Ji, Enzhou Song, and Ruijie Cai, *Information Engineering University*;

Jinglei Tan, *State Key Laboratory of Mathematical Engineering and Advanced Computing*; Shengli Liu, *Information Engineering University*

AVAILABLE MEDIA 

[Show details](#) ▶

Patch-Guided Vulnerability Detection: Extracting Java API Security Rules via Attack-Defense Cross-Analysis

Bofei Chen, Shuang Liao, and Lei Zhang, *Fudan University*; Chibin Zhang and Mathias Payer, *EPFL*; Yuan Zhang, *Fudan University*

AVAILABLE MEDIA 

[Show details](#) ▶

The State of Passkeys: Studying the Adoption and Security of Passkeys on the Web

Louis Jannett, *Ruhr University Bochum*; Andreas Mayer and Maximilian Westers, *Heilbronn University of Applied Sciences*; Vladislav Mladenov, *Ruhr University Bochum*; Christian Mainka, *University of Wuppertal*; Jörg Schwenk, *Ruhr University Bochum*

AVAILABLE MEDIA 

[Show details](#) ▶

StackWarp: Breaking AMD SEV-SNP Integrity via Deterministic Stack-Pointer Manipulation through the CPU's Stack Engine

Ruiyi Zhang, Tristan Horneitz, Daniel Weber, Fabian Thomas, and Michael Schwarz, *CISPA Helmholtz Center for Information Security*

AVAILABLE MEDIA 

[Show details](#) ▶

Leveraging Cryptographic Simulator Synthesis for Formally Verifying the FOO E-Voting Protocol

David Baelde, *Univ Rennes, CNRS, IRISA*; Adrien Koutsos and Justine Sauvage, *Inria*

AVAILABLE MEDIA 

[Show details](#) ▶

The Art of Hide and Seek: Making Pickle-Based Model Supply Chain Poisoning Stealthy Again

Tong Liu and Guozhu Meng, *Institute of Information Engineering, Chinese Academy of Sciences; School of Cyber Security, University of Chinese Academy of Sciences*; Peng Zhou, *Shanghai University*; Zizhuang Deng, *School of Cyber Science and Technology, Shandong University; State Key Laboratory of Cryptography and Digital Economy Security, Shandong University*; Shuaiyin Yao and Kai Chen, *Institute of Information Engineering, Chinese Academy of Sciences; School of Cyber Security, University of Chinese Academy of Sciences*

AVAILABLE MEDIA 

[Show details](#) ▶

B-Privacy: Defining and Enforcing Privacy in Weighted Voting

Samuel Breckenridge, Dani Vilardell, and Andrés Fábrega, *Cornell Tech, IC3*; Amy Zhao, *Ava Labs, IC3*; Patrick McCorry, *Arbitrum Foundation*; Ari Juels, *Cornell Tech, IC3*

AVAILABLE MEDIA 

[Show details ▶](#)

InstrSem: Automatically and Generically Inferring Semantics of (Undocumented) CPU Instructions

Lorenz Hetterich, Fabian Thomas, Tristan Hornetz, and Michael Schwarz, *CISPA Helmholtz Center for Information Security*

AVAILABLE MEDIA 

[Show details ▶](#)

Paper Title Under Embargo

Lorenz Hetterich, Tristan Hornetz, Fabian Thomas, and Michael Schwarz, *CISPA Helmholtz Center for Information Security*

[Show details ▶](#)

Sliding into the Flight Deck's DMs: Practical Message Attacks on CPDLC

Mehdi Ziazi, *ETH Zurich*; Khalid Aleem, *Independent*; Harshad Sathaye, *ETH Zurich*; Martin Strohmeier, *Cyber-Defence Campus, armasuisse Science + Technology*

[Show details ▶](#)

You Know Why, but Still Rely: The Impact of Explainable AI on Trust, Task Load, and Performance in Cybersecurity Decision-Making

Neele Roch, Hannah Sievers, Noé Zufferey, and Verena Zimmermann, *ETH Zurich*

AVAILABLE MEDIA 

[Show details ▶](#)

Silicon Heist: (Ransom) Attacks for Cloud FPGAs via Privilege Escalation

Simon Klix, Felix Hahn, Maik Ender, Nils Albartus, and Christof Paar, *Max Planck Institute for Security and Privacy (MPI-SP)*; Russell Tessier, *University of Massachusetts*

AVAILABLE MEDIA 

[Show details ▶](#)

Bridging Bitcoin to Second Layers via BitVM2

Robin Linus Woll, *Stanford University and ZeroSync Association*; Lukas Aumayr, *University of Edinburgh and Common Prefix*; Zeta Avarikioti, *TU Wien and Common Prefix*; Matteo Maffei, *TU Wien*; Andrea Pelosi, *University of Pisa, University of Camerino, and TU Wien*; Orfeas Stefanos Thyfronitis Litos, *Imperial College London and Common Prefix*; Christos Stefo, *TU Wien*; David Tse, *Stanford University and Byzantine Research*; Alexei Zamyatin, *BOB*

AVAILABLE MEDIA 

[Show details ▶](#)

"Oh, what people would do with my knife?" Navigating the Dual-Use Dilemma in PoC Exploit Development, Disclosure, and Community Dynamics

Arwa Al Alsadi and Lorenz Kustosch, *TU Delft*; Lamya Alowain, *Independent*; Michel Van Eeten and Carlos H. Gañán, *TU Delft*

AVAILABLE MEDIA 

[Show details ▶](#)

Lost in Blockchain Address Misuse: Hidden Cross-Platform Risks and Their Security Impact

Zhenzhe Shao, *Sun Yat-sen University and Zhejiang University*; Jiashuo Zhang, *Peking University*; Zihao Li, *University of Electronic Science and Technology of China and The Hong Kong Polytechnic University*; Daoyuan Wu, *Lingnan University*; Chong Chen and Yiming Shen, *Sun Yat-sen University*; Lingfeng Bao, *Zhejiang University and Hangzhou High-Tech Zone (Binjiang) Institute of Blockchain and Data Security*; Yanlin Wang, *Sun Yat-sen University*; Jiachi Chen, *Zhejiang University and Hangzhou High-Tech Zone (Binjiang) Institute of Blockchain and Data Security*

AVAILABLE MEDIA 

[Show details ▶](#)

DDR-SSE: Duplicated Retrieval of Documents for System-wide Secure Searchable Symmetric Encryption

Zichen Gui, *University of Georgia, USA*; Simon-Philipp Merz and Kenneth G. Paterson, *ETH Zürich, Switzerland*; Sikhar Patranabis, *IBM Research India*

AVAILABLE MEDIA 

[Show details ▶](#)

Concretely Efficient Blind Signatures Based on VOLE-in-the-Head Proofs and the MAYO Trapdoor

Carsten Baum and Marvin Beckmann, *Denmark Technical University*; Ward Beullens, *IBM, Zürich*; Shibam Mukherjee, *Graz University of Technology and Know Center, Graz*; Christian Rechberger, *Graz University of Technology and TACEO, Graz*

AVAILABLE MEDIA 

[Show details ▶](#)

Turn Your Face Into An Attack Surface: Screen Attack Using Facial Reflections in Video Conferencing

Yong Huang, Yanzhao Lu, Mingyang Chen, En Zhang, and Jiazi Li, *Zhengzhou University*; Wanqing Tu, *Durham University*

AVAILABLE MEDIA 

[Show details ▶](#)

PROBE+DETECT+MITIGATE (PDM): Enabling Cloud Tenants to Self-Defend against Microarchitectural Attacks

Arash Daneshmand, *University of British Columbia Okanagan and Concordia University*; Hugo Kermabon-Bobinnec, *Concordia University*; Lingyu Wang, *University of British Columbia Okanagan and Concordia University*; Makan Pourzandi, *Ericsson Security Research, Ericsson Canada*; Suryadipta Majumdar, *Concordia University*; Yosr Jarraya, *Ericsson Security Research, Ericsson Canada*

AVAILABLE MEDIA 

[Show details ▶](#)

ARM MTE Performance in Practice

Taehyun Noh, *University of Texas at Austin*; Yingchen Wang, *University of California Berkeley*; Tal Garfinkel, *Google*; Mahesh Madhav, *Ampere Computing*; Daniel Moghimi, *Google*; Mattan Erez and Shravan Narayan, *University of Texas at Austin*

AVAILABLE MEDIA 

[Show details ▶](#)

Why Johnny Adopts Identity-Based Software Signing: A Usability Case Study of Sigstore
Kelechi G. Kalu, Sofia Okorafor, and Tanmay Singla, *Purdue University*; Sophie Chen, *Carnegie Mellon University*; Santiago Torres-Arias and James C. Davis, *Purdue University*

AVAILABLE MEDIA 

[Show details ▶](#)

Scribe: Low-memory SNARKs via Read-Write Streaming

Anubhav Baweja, Pratyush Mishra, Tushar Mopuri, Karan Newatia, and Steve Wang, *University of Pennsylvania*

AVAILABLE MEDIA 

[Show details ▶](#)

Identifying Provenance of Generative Text-to-Image Models

Anna Yoo Jeong Ha, Wenxin Ding, Stanley Wu, Shawn Shan, Haitao Zheng, and Ben Y. Zhao, *University of Chicago*

AVAILABLE MEDIA 

[Show details ▶](#)

Semantics Over Syntax: Uncovering Pre-Authentication 5G Baseband Vulnerabilities

Qiqing Huang and Xingyu Wang, *University at Buffalo*; Wanda Guo and Guofei Gu, *Texas A&M University*; Hongxin Hu, *University at Buffalo*

AVAILABLE MEDIA 

[Show details ▶](#)

Window-based Membership Inference Attacks Against Fine-tuned Large Language Models

Yuetian Chen, Yuntao Du, and Kaiyuan Zhang, *Purdue University*; Ashish Kundu, *Cisco Research*; Charles Fleming, *Cisco Systems*; Bruno Ribeiro and Ninghui Li, *Purdue University*

AVAILABLE MEDIA 

[Show details ▶](#)

Digital Risks and Coping Practices among Roblox Game Creators

Qiurong Song, Rie Helene (Lindy) Hernandez, Xinning Gui, and Yubo Kou, *The Pennsylvania State University*

AVAILABLE MEDIA 

[Show details ▶](#)

Paper Title Under Embargo

Author list under embargo.

[Show details ▶](#)

SoK: Attack and Defense Landscape of Agentic AI Systems

Juhee Kim, *UC Berkeley and Seoul National University*; Wenbo Guo, *UC Santa Barbara*; Dawn Song, *UC Berkeley*

AVAILABLE MEDIA 

[Show details ▶](#)

Network-Level Prompt and Trait Leakage in Local Research Agents

Hyejun Jeong, Mohammadreza Teymoorianfard, Abhinav Kumar, Amir Houmansadr, and Eugene Bagdasarian, *University of Massachusetts Amherst*

AVAILABLE MEDIA 

[Show details ▶](#)

NOIR: Privacy-Preserving Generation of Code with Open-Source LLMs

Khoa Nguyen, *New Jersey Institute of Technology*; That Khiem Ton, *New Jersey Insititute of Technology*; NhatHai Phan, *New Jersey Institute of Technology*; Issa Khalil, *Hamad Bin Khalifa University*; Khang Tran and Cristian Borcea, *New Jersey Institute of Technology*; Ruoming Jin, *Kent State University*; Abdallah Khreishah, *New Jersey Institute of Technology*; My T. Thai, *University of Florida*

AVAILABLE MEDIA 

[Show details ▶](#)

BADControl: Backdoor Attacks Against Control Systems

Luis Burbano, *University of California, Santa Cruz*; Hampei Sasahara, *Institute of Science Tokyo*; Ruoyu Song and Z. Berkay Celik, *Purdue University*; Alvaro A. Cardenas, *University of California, Santa Cruz*

AVAILABLE MEDIA 

[Show details ▶](#)

Can we estimate privacy vulnerability of individual records? Towards Mitigating Attribute Inference Attacks on ML Models

Ehsanul Kabir and Najrin Sultana, *Pennsylvania State University*; Ninghui Li, *Purdue University*; Shagufta Mehnaz, *Pennsylvania State University*

AVAILABLE MEDIA 

[Show details ▶](#)

Lethe: Purifying Backdoored Large Language Models with Knowledge Dilution

Chen Chen, *Nanyang Technological University*; Yuchen Sun and Jiaxin Gao, *Wuhan University*; Xueluan Gong, *Nanyang Technological University*; Qian Wang, *Wuhan University*; Ziyao Liu, Yongsen Zheng, and Kwok-Yan Lam, *Nanyang Technological University*

AVAILABLE MEDIA 

[Show details ▶](#)

Distributed Synthesis of Differentially Private Tabular Datasets

Yucheng Fu, *University of Virginia*; Tianyao Gu and Elaine Shi, *Carnegie Mellon University*; Tianhao Wang, *University of Virginia*

AVAILABLE MEDIA 

[Show details ▶](#)

Trustworthy and Confidential SBOM Exchange

Eman Abu Ishgair and Chinenye Okafor, *Purdue University*; Marcela S. Melara, *Intel Corporation*; Santiago Torres-Arias, *Purdue University*

AVAILABLE MEDIA 

[Show details ▶](#)

Libra: Pattern-Scheduling Co-Optimization for Cross-Scheme FHE Code Generation over GPGPU

Song Bian, Yintai Sun, Zian Zhao, and Haowen Pan, *Beihang University*; Mingzhe Zhang, *unaffiliated*; Zhenyu Guan, *Beihang University*

AVAILABLE MEDIA 

[Show details](#) ▶

Efficient and High-Accuracy Secure Two-Party Protocols for a Class of Functions with Real-number Inputs

Hao Guo and Zhaoqian Liu, *The Chinese University of Hong Kong, Shenzhen*; Liqiang Peng, *Alibaba Group*; Shuaishuai Li, *Zhongguancun Laboratory, Beijing, China*; Ximing Fu, *The Chinese University of Hong Kong, Shenzhen*; Weiran Liu and Lin Qu, *Alibaba Group*

AVAILABLE MEDIA 

[Show details](#) ▶

CompLeak: Deep Learning Model Compression Exacerbates Privacy Leakage

Na Li, *School of Cyber Science and Engineering, Nanjing University of Science and Technology, China*; Yansong Gao, *School of Cyber Science and Engineering, Southeast University, China*; Hongsheng Hu, *School of Computer Science, Shanghai Jiao Tong University, China*; Boyu Kuang, *School of Cyber Science and Engineering, Nanjing University of Science and Technology, China*; Anmin Fu, *School of Cyber Science and Engineering, Nanjing University of Science and Technology, China*; and *School of Computer Science and Engineering, Nanjing University of Science and Technology, China*

AVAILABLE MEDIA 

[Show details](#) ▶

TAT: Attesting Trajectory Integrity of Industrial Robotic Arms

Chengtao Yao, Chengcheng Zhao, Peng Cheng, and Jiming Chen, *Zhejiang University*

[Show details](#) ▶

VidLeaks: Membership Inference Attacks Against Text-to-Video Models

Li Wang and Wenyu Chen, *Shandong University*; Ning Yu, *Eyeline Labs*; Zheng Li and Shanqing Guo, *Shandong University*

AVAILABLE MEDIA 

[Show details](#) ▶

When Fun Turns Toxic: A First Look at Aggressive Advertising in Mini-games

Pei Chen, Geng Hong, Yicheng Qin, Huazhe Wang, Mengying Wu, and Min Yang, *Fudan University*; Ziru Zhao, Yuanpeng Zhu, and Tao Su, *vivo Mobile Communication Co., Ltd*

AVAILABLE MEDIA 

[Show details](#) ▶

When Memory Becomes a Vulnerability: Towards Multi-turn Jailbreak Attacks against Text-to-Image Generation Systems

Shiqian Zhao, *Nanyang Technological University*; Jiayang Liu, *Nanyang Technological University; and Institute of Science Tokyo, Japan*; Yiming Li, Runyi Hu, and Xiaojun Jia, *Nanyang Technological University*; Wenshu Fan, *University of Electronic Science and Technology of China*; Xiaobao Wu and Xinfeng Li, *Nanyang Technological University*; Jie Zhang, *CFAR and IHPC, Agency for Science,*

*Technology and Research (A*STAR), Singapore; Wei Dong and Tianwei Zhang, Nanyang Technological University; Luu Anh Tuan, Nanyang Technological University and VinUniversity*

AVAILABLE MEDIA 

[Show details ▶](#)

SoK: Security of Cyber-physical Systems Under Intentional Electromagnetic Interference Attacks

Qinhong Jiang, The Hong Kong Polytechnic University; Yan Long, The Hong Kong University of Science and Technology (Guangzhou); Youqian Zhang, The Hong Kong Polytechnic University; Chen Yan and Xiaoyu Ji, Zhejiang University; Xiapu Luo, The Hong Kong Polytechnic University; Kevin Fu, Northeastern University; Jiannong Cao, The Hong Kong Polytechnical University; Wenyan Xu, Zhejiang University

AVAILABLE MEDIA 

[Show details ▶](#)

Fend for Yourself! Backdoor Purification in Federated Graph Learning with an Evolving Knowledge Anchor

Chengcheng Zhu and Yunlong Mao, Nanjing University; Jiale Zhang and Bosen Rao, Yangzhou University; Sheng Zhong, Nanjing University

AVAILABLE MEDIA 

[Show details ▶](#)

InstantOMR: Oblivious Message Retrieval with Low Latency and Optimal Parallelizability

Haofei Liang, Shanghai Jiao Tong University; Zeyu Liu, Yale University; Eran Tromer, Boston University; Xiang Xie, Primus Labs; Yu Yu, Shanghai Jiao Tong University

AVAILABLE MEDIA 

[Show details ▶](#)

Cracks in the Walled Garden: Dissecting the Gray-Market of Unauthorized iOS App Distribution via Ad Hoc Sideloads

Yijing Liu, Yiming Zhang, Baojun Liu, and Haixin Duan, Tsinghua University and BNRist

AVAILABLE MEDIA 

[Show details ▶](#)

ZipPIR: High-throughput Single-server PIR without Client-side Storage

Rasoul Akhavan Mahdavi, Abdulrahman Diao, and Florian Kerschbaum, University of Waterloo

AVAILABLE MEDIA 

[Show details ▶](#)

Provable Secure Steganography Based on Adaptive Dynamic Sampling

Kaiyi Pang and Minhao Bai, Tsinghua University

AVAILABLE MEDIA 

[Show details ▶](#)

Arguzz: Testing zkVMs for Soundness and Completeness Bugs

Christoph Hochrainer, Technische Universität Wien; Valentin Wüstholtz, Diligence Security; Maria Christakis, Technische Universität Wien

AVAILABLE MEDIA 

[Show details ▶](#)

kSFS: Repurposing a Microkernel-like Interface for Fast and Secure In-Kernel Linux File Systems

Dinglan Peng and Pedro Fonseca, *Purdue University*

AVAILABLE MEDIA 

[Show details ▶](#)

Sirens' Whisper: Inaudible Near-Ultrasonic Jailbreaks of Speech-Driven LLMs

Zijian Ling, *Huazhong University of Science and Technology and Tsinghua University*; Pingyi Hu, Xiuyong Gao, and Xiaojing Ma, *Huazhong University of Science and Technology*; Man Zhou, *Huazhong University of Science and Technology and Tsinghua University*; Jun Feng and Songfeng Lu, *Huazhong University of Science and Technology*; Dongmei Zhang and Bin Benjamin Zhu, *Microsoft Corporation*

AVAILABLE MEDIA 

[Show details ▶](#)

Autonomy Comes with Costs: Detecting Denial-of-Service Vulnerabilities Caused by Resource Abusing in LLM-based Agents

Jiaqi Luo, Jiarun Dai, Fengyu Liu, Songyang Peng, Youkun Shi, Tong Bu, and Geng Hong, *Fudan University*; Xudong Pan, *Fudan University and Shanghai Innovation Institute*; Yuan Zhang, *Fudan University*

AVAILABLE MEDIA 

[Show details ▶](#)

Overcoming the Retrieval Barrier: Indirect Prompt Injection in the Wild for LLM Systems

Hongyan Chang, Ergute Bao, Xinjian Luo, and Ting Yu, *Mohamed bin Zayed University of Artificial Intelligence*

AVAILABLE MEDIA 

[Show details ▶](#)

Inference Attacks Against Graph Generative Diffusion Models

Xiuling Wang and Xin Huang, *Hong Kong Baptist University*; Guibo Luo, *Peking University*; Jianliang Xu, *Hong Kong Baptist University*

AVAILABLE MEDIA 

[Show details ▶](#)

Unlocking the True Potential of Decryption Failure Oracles: A Hybrid Adaptive-LDPC Attack on ML-KEM Using Imperfect Oracles

Qian Guo, Denis Nabokov, and Thomas Johansson, *Lund University*

AVAILABLE MEDIA 

[Show details ▶](#)

vCause: Efficient and Verifiable Causality Analysis for Cloud-based Endpoint Auditing

Qiyang Song, Qihang Zhou, Xiaoqi Jia, and Zhenyu Song, *Institute of Information Engineering, Chinese Academy of Sciences; and School of Cyber Security, University of Chinese Academy of Sciences*; Wenbo Jiang, *University of Electronic Science and Technology of China*; Heqing Huang,

Independent Researcher; Yong Liu, Qi An Xin Technology Group Inc.; Dan Meng, Institute of Information Engineering, Chinese Academy of Sciences; and School of Cyber Security, University of Chinese Academy of Sciences

AVAILABLE MEDIA 

[Show details ▶](#)

CombiSan: Unifying Software Sanitizers for Comprehensive Fuzzing

Matteo Marini, *Sapienza University of Rome*; Floris Gorter, *Vrije Universiteit Amsterdam*; Daniele Cono D'Elia, *Sapienza University of Rome*; Cristiano Giuffrida, *Vrije Universiteit Amsterdam*

AVAILABLE MEDIA 

[Show details ▶](#)

KernelRCA: Facilitating Root Cause Analysis of Memory Corruptions in Linux Kernel with Contextual Causality Chain

Kangzheng Gu, Yifan Zhang, Yuan Zhang, and Min Yang, *Fudan University*

AVAILABLE MEDIA 

[Show details ▶](#)

Sy-FAR: Symmetry-based Fair Adversarial Robustness

Haneen Najjar, Eyal Ronen, and Mahmood Sharif, *Tel Aviv University*

AVAILABLE MEDIA 

[Show details ▶](#)

Ajax: Fast Threshold Fully Homomorphic Encryption without Noise Flooding

Zhenkai Hu, *Shanghai Jiao Tong University and State Key Laboratory of Cryptology*; Haofei Liang, *Shanghai Jiao Tong University*; Xiao Wang, *Northwestern University*; Xiang Xie, *East China Normal University and Primus Labs*; Kang Yang, *State Key Laboratory of Cryptology*; Yu Yu, *Shanghai Jiao Tong University*; Wenhao Zhang, *Northwestern University*

AVAILABLE MEDIA 

[Show details ▶](#)

FirmReBugger: A Benchmark Framework for Monolithic Firmware Fuzzers

Mathew Duong, Michael Chessner, and Guy Farrelly, *University of Adelaide*; Surya Nepal, *Data61 CSIRO*; Damith C. Ranasinghe, *University of Adelaide*

AVAILABLE MEDIA 

[Show details ▶](#)

Differential Trust: Dynamic Multi-Authority Anonymous Credentials with Epoch-Weighted Updates

Chen Li, *Tianjin University*; Jianting Ning, *Zhejiang Sci-Tech University*; Xiulong Liu, *Tianjin University*; Yulin Liu, *Wuhan University*

AVAILABLE MEDIA 

[Show details ▶](#)

From Texts to Rules: Generating Sigma Rules with Large Language Models from Cyber Threat Reports

Yongxin Cai, *Guangzhou University*; Jing Qiu, *Guangzhou University and Pengcheng Laboratory*; Qingming Li, *Zhejiang University*; Du Cheng, *Tsinghua University*; Lei Chen, *Hong Kong University of Science and Technology*

AVAILABLE MEDIA 

[Show details](#) ▶

Static Detection of TOCTOU Bugs Caused by Kernel Races

Gui-Dong Han, Jia-Ju Bai, Qiu-Ji Chen, and Jiqiang Lu, *Beihang University*

AVAILABLE MEDIA 

[Show details](#) ▶

Paper Title Under Embargo

Benedict Schlüter, Christoph Wech, and Shweta Shinde, *ETH Zurich*

[Show details](#) ▶

OS-Sanitizer: System-wide Latent Defect Inference in Linux Applications

Addison Crump, Sahil Sihag, Florian Bauckholt, and Keno Hassler, *CISPA Helmholtz Center for Information Security*; Thorsten Holz, *Max Planck Institute for Security and Privacy*

AVAILABLE MEDIA 

[Show details](#) ▶

SafeFFI: Efficient Sanitization at the Boundary Between Safe and Unsafe Code in Rust and Mixed-Language Applications

Oliver Braunsdorf and Tim Lange, *Ludwig-Maximilians-Universität München*; Konrad Hohentanner and Julian Horsch, *Fraunhofer AISEC*; Johannes Kinder, *Ludwig-Maximilians-Universität München*

AVAILABLE MEDIA 

[Show details](#) ▶

GateBreaker: Gate-Guided Attacks on Mixture-of-Expert LLMs

Lichao Wu, Sasha Behrouzi, and Mohamadreza Rostami, *Technical University of Darmstadt*; Stjepan Picek, *University of Zagreb and Radboud University*; Ahmad-Reza Sadeghi, *Technical University of Darmstadt*

AVAILABLE MEDIA 

[Show details](#) ▶

Bridges to Self: Silent Web-to-App Tracking on Mobile via Localhost

Tim Vlummens, *COSIC, KU Leuven*; Aniketh Girish and Nipuna Weerasekara, *IMDEA Networks Institute*; Frederik Zuiderveen Borgesius and Gunes Acar, *Radboud University*; Narseo Vallina-Rodriguez, *IMDEA Networks Institute*

AVAILABLE MEDIA 

[Show details](#) ▶

WILD Attack: Stealthy Undermining of Wi-Fi-Based Geolocation Through Remote Crowdsourced Data Injection

Changjia Zhu, Xiao Han, Parush Gera, Zhuo Lu, Tempestt Neal, and Yao Liu, *University of South Florida*

AVAILABLE MEDIA 

[Show details ▶](#)

When Updates Backfire: A Black-Box Security Analysis of Desktop Software Update Mechanisms

Jie Wan, Pengcheng Xia, and Haoyu Wang, *Huazhong University of Science and Technology*

AVAILABLE MEDIA 

[Show details ▶](#)

TopFeaRe: Locating Critical State of Adversarial Resilience for Graphs Regarding Topology-Feature Entanglement

Xinxin Fan, *State Key Laboratory of AI Safety, Institute of Computing Technology, Chinese Academy of Sciences; and University of Chinese Academy of Sciences*; Wenxiong Chen, *Dalian University of Technology; and State Key Laboratory of AI Safety, Institute of Computing Technology, Chinese Academy of Sciences*; Quanliang Jing, *Institute of Computing Technology*; Chi Lin, *Dalian University of Technology*; Shaoye Luo, *State Key Laboratory of AI Safety, Institute of Computing Technology, Chinese Academy of Sciences; and University of Chinese Academy of Sciences*; Wenbo Song, *Dalian University of Technology; and State Key Laboratory of AI Safety, Institute of Computing Technology, Chinese Academy of Sciences*; Yunfeng Lu, *Beihang University*

AVAILABLE MEDIA 

[Show details ▶](#)

BatchBoot: Fast Batched Bootstrapping for TFHE scheme and Practical Applications

Zhihao Li, *Ant Digital Technologies, Ant Group*; Hongyu Wang, *Shanxi University*; Yuan Zhao and Lichun Li, *Ant Digital Technologies, Ant Group*; Zhiwei Wang, *State Key Laboratory of Cyberspace Security Defense, Institute of Information Engineering, CAS*; Jiaxing He, Changzheng Wei, and Ying Yan, *Ant Digital Technologies, Ant Group*; Lifeng Guo, *Shanxi University*

AVAILABLE MEDIA 

[Show details ▶](#)

The Adverse Effects of Omitting Records in Differential Privacy: How Sampling and Suppression Degrade the Privacy–Utility Tradeoff

Àlex Miranda-Pascual, *Karlsruhe Institute of Technology and Universitat Politècnica de Catalunya*; Javier Parra-Arnau, *Universitat Politècnica de Catalunya*; Thorsten Strufe, *Karlsruhe Institute of Technology*

AVAILABLE MEDIA 

[Show details ▶](#)

VeCT: Secure and Efficient Constant-Time Code Rewriting with Vector Extensions

Qisheng Jiang and Danfeng Zhang, *Duke University*

AVAILABLE MEDIA 

[Show details ▶](#)

Memclave: Secure In-memory Enclave for Untrusted Hosts

Amit Choudhari, *CISPA Helmholtz Center for Information Security*; Fabian van Rissenbeck, *Technische Universität Dortmund*; Christian Rossow, *CISPA Helmholtz Center for Information Security*

AVAILABLE MEDIA 

[Show details ▶](#)

Cracking Federated Privacy: Initialization-Resilient Gradient Inversion with Fine-Grained Reconstruction

Kaiming Zhu, Jinsheng Yang, Siyang Guo, Huaqian Qin, Taiyu Wang, Junbo Wang, Yuhong Nan, and Zibin Zheng, *Sun Yat-sen University, P.R. China*

AVAILABLE MEDIA 

[Show details ▶](#)

Estimating the Amount of Script-generated Traffic in a Mixture

Cormac Herley, *Microsoft Research*

[Show details ▶](#)

Quorus: Efficient, Scalable Threshold ML-DSA Signatures from MPC

Alexander Bienstock, Leo de Castro, Daniel Escudero, Antigoni Polychroniadou, and Akira Takahashi, *J.P. Morgan AlgoCRYPT CoE and J.P. Morgan AI Research*

AVAILABLE MEDIA 

[Show details ▶](#)

Chameleon Channels: Measuring YouTube Accounts Repurposed for Deception and Profit

Alejandro Cuevas, *Carnegie Mellon University*; Manoel Horta Ribeiro, *Princeton University*; Nicolas Christin, *Carnegie Mellon University*

AVAILABLE MEDIA 

[Show details ▶](#)

Khost: KVM-based Near Native MCU Firmware Rehosting

Chunlin Wang, Yicheng Yang, Yuan Zhang, Haoyu Xiao, Yifan Zhang, and Jiarun Dai, *Fudan University*

AVAILABLE MEDIA 

[Show details ▶](#)

Attesting Model Lineage by Consisted Knowledge Evolution with Fine-Tuning Trajectory

Zhuoyi Shang, Jiasen Li, and Pengzhen Chen, *Institute of Information Engineering, Chinese Academy of Sciences; School of Cyber Security, University of Chinese Academy of Sciences; and Key Laboratory of Cyberspace Security Defense*; Yanwei Liu, *Institute of Information Engineering, Chinese Academy of Sciences; and Key Laboratory of Cyberspace Security Defense*; Xiaoyan Gu, *Institute of Information Engineering, Chinese Academy of Sciences; School of Cyber Security, University of Chinese Academy of Sciences; and Key Laboratory of Cyberspace Security Defense*; Weiping Wang, *Institute of Information Engineering, Chinese Academy of Sciences*

AVAILABLE MEDIA 

[Show details ▶](#)

SONIC: Concurrent Oblivious RAM & Data Structures for Low-Latency and High-Throughput

Nihal Talur and Ioannis Demertzis, *UC Santa Cruz*

[Show details ▶](#)

E2E-AKMA: An End-to-End Secure and Privacy-Enhancing AKMA Protocol Against the Anchor Function Compromise

Yueming Li, *Institute of Software, Chinese Academy of Sciences; and University of Chinese Academy of Sciences*; Long Chen, *Institute of Software, Chinese Academy of Sciences; and Key Laboratory of System Software, Chinese Academy of Sciences*; Qianwen Gao, *Institute of Software, Chinese Academy of Sciences; and University of Chinese Academy of Sciences*; Zhenfeng Zhang, *Institute of Software, Chinese Academy of Sciences; and Key Laboratory of System Software, Chinese Academy of Sciences*

AVAILABLE MEDIA 

[Show details ▶](#)

Adversarial Patch EXterminator: Zero-Shot and Patch-Agnostic Defense Framework Against Adversarial Patch Attacks

Jiayimei Wang, *City University of Hong Kong*; Tao Ni, *King Abdullah University of Science and Technology*; Guowen Xu, *University of Electronic Science and Technology of China*; Qingchuan Zhao and Cong Wang, *City University of Hong Kong*

AVAILABLE MEDIA 

[Show details ▶](#)

Quantifying Large Language Model Attacks Through the Lens of Model Cognition

Xiuming Liu, Chaoxiang He, Xuanran Yu, Jichen Chai, Feiyue Xu, Sheng Hang, and Hanqing Hu, *Shanghai Jiao Tong University*; Bin Benjamin Zhu, *Microsoft Corporation*; Hongsheng Hu, Shifeng Sun, Dawu Gu, and Shuo Wang, *Shanghai Jiao Tong University*

AVAILABLE MEDIA 

[Show details ▶](#)

ATTEND

Registration Information

Registration Discounts

Grant Opportunities

Venue, Hotel, and Travel

CYCLE 1 ACCEPTED PAPERS

PARTICIPATE

Call for Papers

Call for Artifacts

Enigma Track Call for Participation

Instructions for Presenters and Authors

SPONSORS

ABOUT

Symposium Organizers

Past Symposia

Conference Policies

Code of Conduct

Questions

[Privacy Policy](#) [Contact Us](#)

© USENIX 2026

EIN 13-3055038